AdS-GNN

We introduce **AdS-GNN**: a **graph neural network** that uses AdS space to organize the computation.

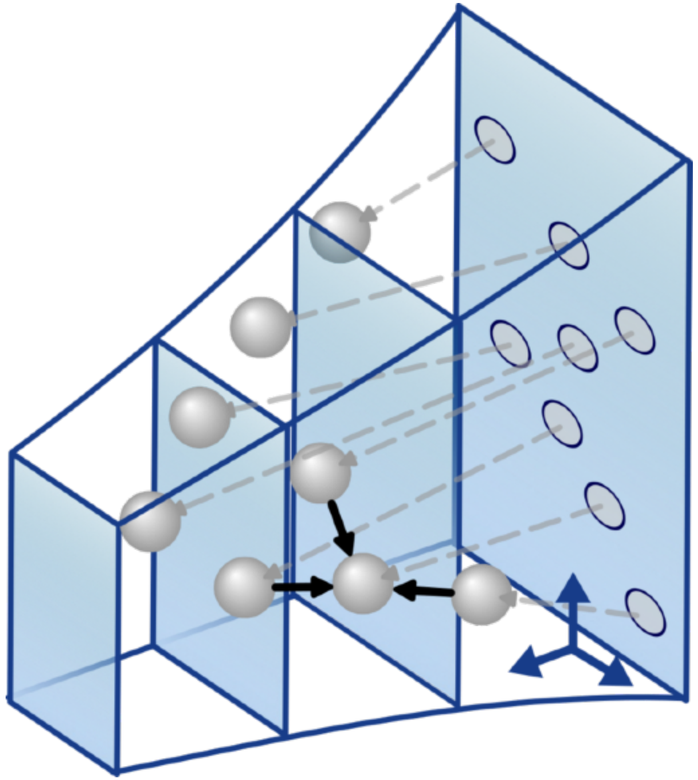
Idea:

1. Take an input point cloud on $\mathbb{R}^2$ to be on the boundary of AdS.

2. **Lift the data** into the interior of AdS in a conformally covariant way.

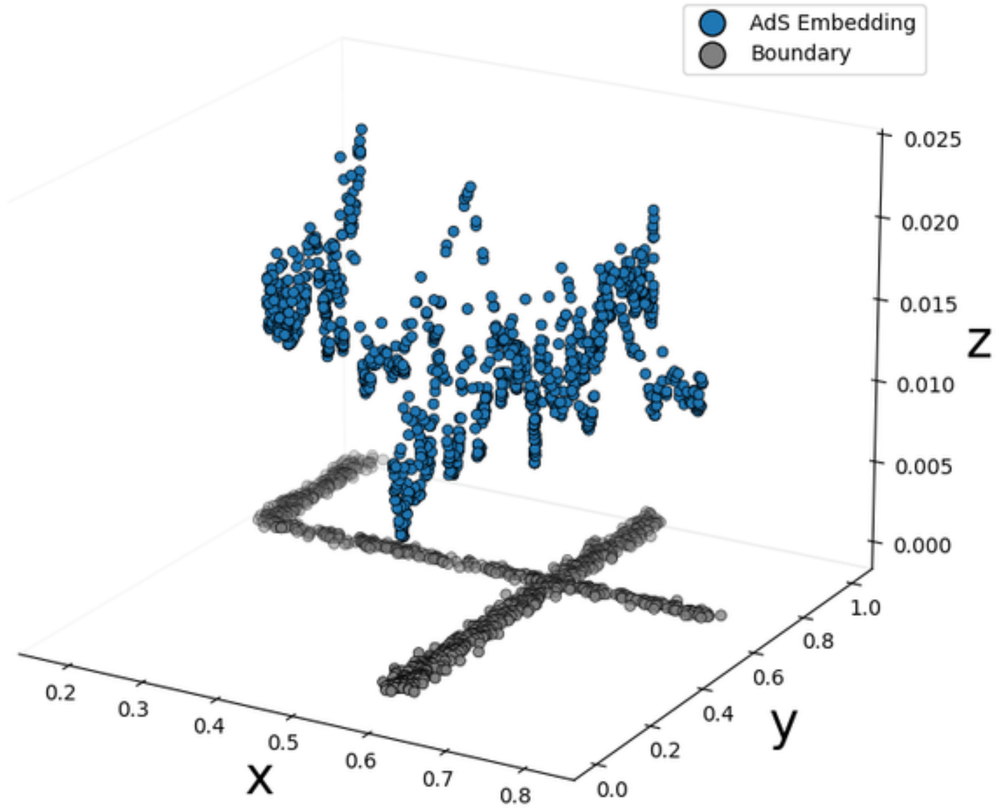
3. **Do bulk operations** in a way that respects AdS isometries using known equivariant neural network tools generalized to AdS.

4. Read out the answer at the end.



AdS-GNN: Lifting

$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$



Need to assign a z_i value to each point.

This can be interpreted as a **length scale**: obtain this by averaging distance from its *k* nearest neighbours in a covariant way.

Points in low-density regions end up further in the interior.